Notation

The literatures this book spans grew up separately and their symbols collide; these are the conventions used here, followed by the symbols that carry more than one meaning.

PRICES, RATES, AND TIME

t	the present, or a general date
T	maturity, expiration, or settlement date
τ	accrual fraction between two dates, in years
S, S_0, S_T	spot price of the underlying: generally, now, at maturity
F, F_0	forward or futures price
K	strike price, or the delivery price of a forward
r	continuously compounded risk-free rate
q	dividend yield; the foreign risk-free rate for a currency
y	convenience yield on a commodity
L, N	notional principal

OPTIONS AND VOLATILITY

c, p	prices of a European call and put (Chapters 11, 15, 16)
C, P	prices of an American call and put (Chapter 11); P standing alone is never a bond price
σ	volatility; implied volatility where the context is a quote
$N(x)$	standard normal cumulative distribution function
$N'(x)$	standard normal density, $e^{-x^2/2}/\sqrt{2\pi}$; written ϕ or φ in Chapters 24, 25, 29 and 36
d_1, d_2	the arguments of the Black–Scholes and Black formulas
Δ, Γ, Θ, $\mathcal{V}$	delta, gamma, theta, vega
H	barrier level
κ	speed of mean reversion of variance in a stochastic volatility model
H (Chs. 14, 21)	Hurst parameter of fractional Brownian motion, and of a rough volatility model

TERM STRUCTURE

$P(t,T)$	price at t of a zero-coupon bond maturing at T
$A(t,T)$, $B(t,T)$	the coefficients of an affine bond price, $P = A\,e^{-Br}$
$B(t)$	value at t of one unit invested from time 0 at the overnight rate used for discounting; the accumulation account
A	annuity, or PV01 leg, of a swap
a	speed of mean reversion of the short rate
Q	the risk-neutral measure
Q^T, Q^N	the T-forward measure and the measure induced by numeraire N
$\mathbb{E}$	expectation, with the measure carried as a superscript
Λ	a Radon–Nikodym derivative: dQ/dP in Chapter 28, and the ratio of two forward measures in Chapter 30; Λ_t is its restriction to $\mathcal{F}_t$, the density process

CREDIT

λ	hazard rate, or default intensity
τ	time of default
R	recovery rate
A	risky annuity: the present value of one basis point paid until default
ρ	default correlation

RISK

α	confidence level of a risk measure
μ	drift, or mean
ρ	correlation
β	sensitivity to a common factor

SYMBOLS WITH MORE THAN ONE MEANING

- ρ: a *correlation* nearly everywhere (including price against variance in a stochastic volatility model), and the Greek *rho* in Chapter 18.
- H: a *CSA threshold* in Chapter 9, a *barrier level* in Chapter 34, the *Hurst parameter* in Chapters 14 and 21, a *futures price* in Chapter 28; no chapter uses two at once.
- B: the *accumulation account* $B(t)$ (Chapters 28 and 30) and an *affine coefficient* $B(t,T)$ (Chapter 31); the arguments separate them.
- τ: an *accrual fraction* in the rates chapters, the *time of default* in Chapters 9, 24 and 25.
- A: an *annuity* in the swap chapters, a *risky annuity* in the credit chapters; the same construction, stopping or not at default.

The normal distribution function is $N(\cdot)$, not $\Phi(\cdot)$, except in Chapter 36 and one formula of Chapter 28, where N is a notional. $P(t,T)$ is always a bond price; P alone is the American put of Chapter 11.

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